

FPGA-Based Number Comparison Game

Lab Report

Department of Computer Science & Engineering

1. Introduction

In this lab, we designed a simple FPGA-based game where two input numbers, A and B, are compared. The result is displayed on a 7-segment display as “HI”, “LO”, or the final score. The game updates the score with each button press and ends when A equals B. This project introduces key FPGA design concepts.

2. Objectives

- Design game logic using Verilog HDL.
 - Implement button debouncing and one-pulse generation.
 - Learn two-digit 7-segment display multiplexing.
 - Write modular, synthesizable, and clean FPGA code.
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3. System Overview

The project is divided into five Verilog modules:

1. `compare_game_top` – Main game logic, score counter, and comparison control.
 2. `debounce_onepulse` – Removes button bounce and generates a clean pulse.
 3. `sevenseg_hex` – Converts numeric digits (0–9, A–F) into 7-segment patterns.
 4. `sevenseg_alpha` – Generates patterns for letters H, I, L, O.
 5. `sevenseg_mux2` – Multiplexes two digits for stable display.
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4. Working Principle

Button Press Operation

When the button `try_btn` is pressed:

- The debounce module creates a clean pulse.
- A and B are compared:

Condition	Display	Action
$A > B$	HI	<code>score++</code>
$A < B$	LO	<code>score++</code>
$A == B$	score	<code>game_over = 1</code>

Display Logic

- Game active → Shows “HI” or “LO”.
- Game over → Shows final score (00–99) on two 7-segment displays.

5. Hardware Used

- FPGA Board (Basys-3 / Nexys-A7 / DE1-SoC)
- On-board 50–100 MHz clock
- Push button
- Two-digit 7-segment display

6. Button Debouncing

- Debounce time: 10 ms.
- Uses synchronizer flip-flops and rising-edge detection.
- Ensures one action per button press.

7. 7-Segment Multiplexing

- Digits switch rapidly (≈ 1 kHz each).
 - Creates persistence of vision effect.
 - `an[0]` = right digit, `an[1]` = left digit.
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8. Results

- ✓ "HI" displayed correctly when $A > B$.
 - ✓ "LO" displayed correctly when $A < B$.
 - ✓ Score increments properly.
 - ✓ Game ends and shows final score when $A == B$.
 - ✓ No button bounce issues.
 - ✓ Stable display with no flickering.
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9. Discussion

- Modular design simplified debugging.
 - Debounce circuit worked perfectly.
 - Displaying letters was challenging but solved with `sevenseg_alpha`.
 - Key concepts learned: synchronous logic, timing constraints, display multiplexing.
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10. Conclusion

This lab successfully implemented an FPGA-based comparison game. Students gained practical experience in:

- FPGA design flow
- Button debouncing
- 7-segment multiplexing
- Verilog modular coding

Future improvements:

- Add random number generator
 - Add game timer
 - Add buzzer sound
 - Use 4-digit display for more features
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11. Appendix

Verilog source code included in project files.